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Studierichting: Informatica

Oefeningenreeks 1B nr. 43.20

$$(1+i)z^2 + (3-7i)z - 10 = 0$$

$$\Delta = (3-7i)^2 - 4(1+i)(-10)$$

$$= 9 - 42i + 49i^2 + 40 + 40i$$

$$= 9 - 42i - 49 + 40 + 40i$$

$$= -2i$$

$$-2i = (a+bi)^2 = a^2 - b^2 + 2abi$$

$$\Rightarrow \begin{cases} a^2 - b^2 = 0 \\ ab = -1 \end{cases}$$

$$\Rightarrow \begin{cases} a^2 = b^2 \\ b = \frac{-1}{a} \end{cases}$$

$$\Rightarrow \begin{cases} a^2 = \frac{1}{a^2} \\ b = \frac{-1}{a} \end{cases}$$

$$\Rightarrow \begin{cases} a^4 = 1 \\ b = \frac{-1}{a} \end{cases}$$

$$\Rightarrow a = 1, b = -1 \text{ of } a = -1, b = 1$$

$$\Rightarrow z_1 = \frac{-(3-7i) + (1-i)}{2(1+i)} = \frac{-2+6i}{2+2i} = \frac{-1+3i}{1+i}$$

$$= \frac{(-1+3i)(1-i)}{(1+i)(1-i)} = \frac{2+4i}{2} = 1+2i$$

$$\Rightarrow z_2 = \frac{-(3-7i) - (1-i)}{2(1+i)} = \frac{-4+8i}{2+2i} = \frac{-2+4i}{1+i}$$

$$= \frac{(-2+4i)(1-i)}{(1+i)(1-i)} = \frac{2+6i}{2} = 1+3i$$

$$\Rightarrow \boxed{z_1 = 1+2i, z_2 = 1+3i}$$

References